HAYFAE TAOUD EL HASSANI

Mechanical Engineer – Applicant for Doctoral Research in AI & Computer Vision



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Mechanical engineer specialised in numerical modelling and simulation, with a Master 2 in Mechanical Modelling for Energy and Environment (Strasbourg) and an engineering degree in Automotive Engineering (UIR Rabat). My final-year research project at the ICube laboratory (CNRS / INSA Strasbourg) gave me hands-on experience in multiphysics modelling, experimental validation and infrared thermography of lithium-ion batteries, resulting in two IEEE publications. Passionate about artificial intelligence and machine learning, I am applying for a PhD to move into computer vision, anomaly detection and multimodal learning, building on a strong applied-mathematics and simulation background.

Python – C / C++ – MATLAB – TensorFlow – Applied Mathematics – Experimental Validation – Predictive Indicators

RESEARCH INTERESTS & PHD PROJECT

- **AI & Computer Vision :** Strong motivation for machine learning, practised in Python (TensorFlow) during my internships; keen to build solid foundations in image processing and deep learning for vision (PyTorch).
- **Research Topics of Interest :** Anomaly detection from visual and thermal data, vision-language and multimodal learning, 3D understanding and generation, and real-time AI deployment.
- **Simulation & Applied Mathematics :** Multiphysics and numerical modelling (COMSOL, finite-element methods) validated against experimental data; physics-based simulation as a source of synthetic data for learning.

PROFESSIONAL EXPERIENCE

Research Engineer – Thermal Modelling of Li-ion Batteries (THERMOBATT project)

ICube (CNRS / INSA Strasbourg) | Final-year project | Strasbourg, France Feb 2026 – Jul 2026

- **Multiphysics Modelling :** Developed and validated a 3D electro-thermal COMSOL model of cylindrical lithium-ion cells (LFP and NMC chemistries), dedicated to the prevention of thermal runaway.
- **Experimental Validation :** Correlated simulation and measurements using an instrumented test bench and infrared thermography (thermal image acquisition) to monitor the thermal behaviour of the cells.
- **Predictive Indicators :** Simulated critical scenarios and extracted predictive indicators to anticipate thermal runaway situations.
- **Literature Review & Publications :** Structured review of thermal runaway mechanisms and modelling approaches; results published in two IEEE papers (VPPC 2026, EUVIP 2026) and in the final-year thesis.

Engineering Intern – Reduction of Electrical Failures in Static Zone 2

Groupe Renault | Internship | Tangier, Morocco Jun 2025 – Jul 2025

- **Diagnosis & Root-Cause Analysis :** Field diagnosis and root-cause analysis of recurrent electrical failures in Static Zone 2 of an automotive production line.
- **Corrective Actions :** Proposed and implemented corrective actions together with the maintenance and production teams.
- **Results & Performance Tracking :** Reduced the electrical defect rate by 85% in 8 weeks; tracked performance indicators to sustain the gains.

Engineering Intern – Implementation of the ISO 50001 Standard

TE Connectivity | Internship | Tangier, Morocco Apr 2023 - Jun 2023

- **Energy Audit :** Carried out the initial energy audit of the site and prioritised the main energy-consuming areas.
- **ISO 50001 Deployment :** Contributed to setting up the energy management system according to ISO 50001.
- **Data Analysis & Reporting :** Collected and analysed consumption data; wrote technical reports and follow-up summaries.

PUBLICATIONS

- **IEEE VPPC 2026:** *Three-Dimensional Electro-Thermal Modeling and Experimental Validation of Cylindrical Lithium Iron Phosphate Cells for Advanced Battery Thermal Management*
- **EUVIP 2026:** *Comparative Experimental Validation of a Three-Dimensional Electro-Thermal Model for Cylindrical LFP and NMC Lithium-Ion Cells*

TECHNICAL SKILLS

- **Programming :** Python, C, C++, MATLAB, R, MySQL
- **AI / Machine Learning :** Python (TensorFlow), data analysis, predictive indicators; to develop: PyTorch, computer vision
- **Modelling / Simulation :** COMSOL (main tool), ANSYS, ABAQUS, STAR-CCM+, LS-DYNA, CATIA
- **Energy / Systems :** thermal management, Li-ion batteries, EV / HEV systems, renewable energy, control basics

EDUCATION

Master 2 – Mechanical Modelling for Energy and Environment

Faculty of Physics and Engineering, Strasbourg | 2025 - 2026

- Multiphysics modelling (COMSOL)
- Thermal management & Li-ion batteries
- Numerical simulation (ANSYS, ABAQUS)

Engineering Degree – Automotive Engineering

International University of Rabat | 2023 - 2026

- EV / HEV systems
- CFD & fluid mechanics (STAR-CCM+)
- Structures, vibrations & crash (ABAQUS, LS-DYNA)

Bachelor – Renewable Energies

Faculty of Sciences and Techniques, Tangier | 2022 - 2023

- Renewable energies (2022 - 2023)
- DEUST – University Diploma in Sciences and Techniques (2019 - 2022)

ACADEMIC PROJECTS

- **Unsteady CFD – OAT15 Airfoil :** Unsteady simulation around the OAT15 airfoil with STAR-CCM+ and ANSYS: setup, meshing, post-processing and analysis of results.
- **Copper Tensile Test :** ABAQUS simulation: boundary conditions, extraction of stress–strain curves and interpretation.
- **Vibration Analysis :** ANSYS study of a radial engine and of the vibrations of a shock absorber.

LANGUAGES & ADDITIONAL INFORMATION

- **English:** C2 ; **French:** C2 ; **German:** B2 ; **Spanish:** A1
- **Mobility:** national & international ; **Driving licence:** B ; **LinkedIn:** linkedin.com/in/hayfae-taoud-el-hassani